

CUSTOMER CHURN PREDICTION

HELPING SYRIATEL KEEP MORE CUSTOMERS

Name: Chris Mau Karagu
Phase 3 Project

Overview

- SyriaTel is losing customers every month.
- Losing customers means losing revenue.
- It is much cheaper to keep current customers than to find new ones.
- The goal is to create a simple tool that predicts which customers are likely to leave soon.

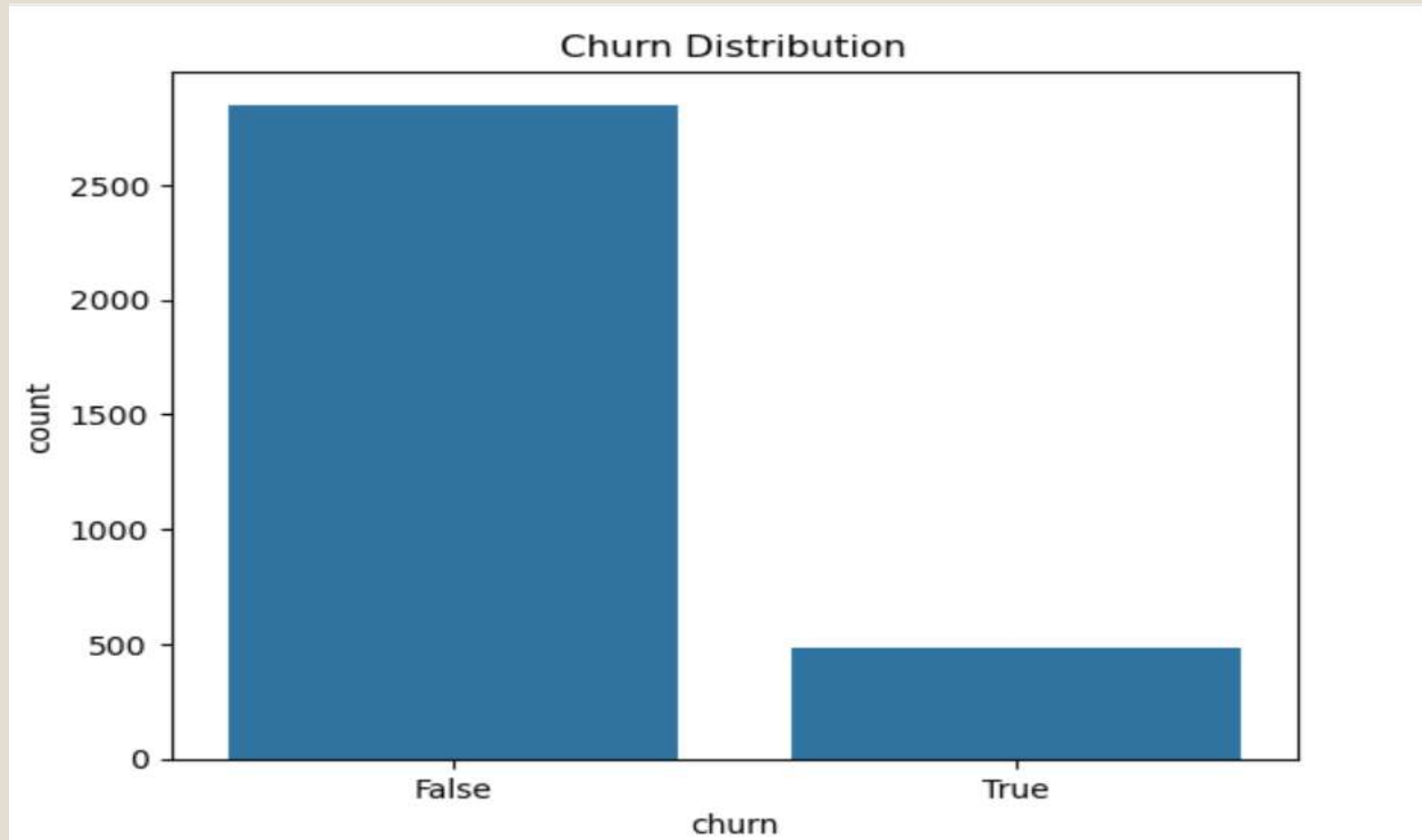
Business Problem

- Customer churn is a big challenge for SyriaTel.
- We want to identify at-risk customers before they leave.
- Early action can help us retain them.
- Success = Reducing customer loss and protecting company income.

Data Understanding

- I analyzed data from 3,333 customers.
- Information included: call minutes, charges, international plan, voicemail plan, customer service calls, etc.
- The data showed clear patterns between customer behavior and churn.
- Key fact: Only 14.5% of customers churned (483 left, 2,850 stayed).
- This is well observed on the next slide.

Data Understanding



Modelling

- I built a prediction tool that answers the question: “Is this customer likely to leave or stay?”
- I used classification as it sorts customers into two groups: Likely to stay and Likely to churn.
- This is useful because instead of waiting for customers to cancel, we can know in advance and take action to keep them.
- I tested different approaches and chose the best one for our business needs.

Evaluation

- I started with Logistic Regression. It was simple but missed many churners.
- I then tried a balanced version of it. It found more churners but gave too many false alarms.
- Finally, I used a Decision Tree. It performed best overall.

Model Results

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	precision	recall	f1-score	support
0	0.94	0.96	0.95	570
1	0.76	0.65	0.70	97
accuracy			0.92	667
macro avg	0.85	0.81	0.83	667
weighted avg	0.92	0.92	0.92	667

Model Results

- Decision Tree Results:
 - ❖ Correctly finds 65% of customers who leave.
 - ❖ When it flags someone, it is right 76% of the time.
 - ❖ Overall accuracy: 92%
- This is good for business. We can catch many at-risk customers without wasting effort on too many wrong predictions.

What We Learned

- The model taught us important things:
 - ❖ Many customer service calls resulted in higher chance of leaving.
 - ❖ International plan users churn more.
 - ❖ Voice mail users usually stay longer.
- These patterns help us know where to focus.

Recommendations

- To reduce churn, SyriaTel should:
 - ❖ Improve customer service.
 - ❖ Check international plan pricing.
 - ❖ Promote voicemail services.
 - ❖ Use the tool every month to find at-risk customers and offer help.

Next Steps

- Use the model every month.
- Combine with special offers.
- Measure how many customers we keep.

Thank You!